

Vector Angle Calculation

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Question

If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

Theoretical Identities

The solution uses two key identities. First, **Lagrange's identity**, which relates the cross product and the dot product (represented by transpose notation):

$$|\mathbf{a} \times \mathbf{b}|^2 + (\mathbf{a}^\top \mathbf{b})^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 \quad (1)$$

Second, the definition of the dot product using matrix transpose notation:

$$\mathbf{a}^\top \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos(\theta) \quad (2)$$

We use (??) to find the value of $\mathbf{a}^\top \mathbf{b}$, and then (??) to find the angle θ .

Solution using Matrices

Let the vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We represent it as a column matrix:

$$\mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (3)$$

The squared magnitude is found by the matrix product $\mathbf{c}^T \mathbf{c}$:

$$\begin{aligned} \|\mathbf{c}\|^2 &= \begin{pmatrix} 3 & 2 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \\ &= (3)(3) + (2)(2) + (6)(6) = 9 + 4 + 36 = 49 \end{aligned}$$

So, the magnitude is $|\mathbf{c}| = \sqrt{49} = 7$.

Finding the Angle

Using Lagrange's identity with $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$, and $|\mathbf{a} \times \mathbf{b}| = 7$:

$$7^2 + (\mathbf{a}^\top \mathbf{b})^2 = (2)^2(7)^2$$

$$49 + (\mathbf{a}^\top \mathbf{b})^2 = 4 \cdot 49 = 196$$

$$(\mathbf{a}^\top \mathbf{b})^2 = 196 - 49 = 147$$

Taking the positive root, we find $\mathbf{a}^\top \mathbf{b} = \sqrt{147} = 7\sqrt{3}$. Now, using the transpose notation for the dot product:

$$\mathbf{a}^\top \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(\theta)$$

$$7\sqrt{3} = (2)(7) \cos(\theta)$$

$$\cos(\theta) = \frac{7\sqrt{3}}{14} = \frac{\sqrt{3}}{2}$$

Therefore, the angle is $\theta = 30^\circ$ or $\frac{\pi}{6}$ radians.

Solution 1: Pure C Code

```
#include <stdio.h>
#include <math.h>

#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

// Calculates angle using Lagrange's identity and the dot product
double angle_from_cross(double a, double b, double x, double y,
    double z) {
    double cross_mag_sq = x*x + y*y + z*z;
    double dot_prod_sq = (a*a * b*b) - cross_mag_sq;
    double dot_prod = sqrt(dot_prod_sq);
    double cos_theta = dot_prod / (a * b);
    double theta_rad = acos(cos_theta);
    return theta_rad * 180.0 / M_PI;
}

int main() {
```

Solution 2: Python + C (The C Library)

```
#include <math.h>

#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

// Function to be compiled into a shared library (e.g., vector.so)
double angle_from_cross_c(double a, double b, double x, double y,
    double z) {
    double cross_mag_sq = x*x + y*y + z*z;
    double dot_prod_sq = (a*a * b*b) - cross_mag_sq;
    double dot_prod = sqrt(dot_prod_sq);
    double cos_theta = dot_prod / (a * b);
    double theta_rad = acos(cos_theta);
    return theta_rad * 180.0 / M_PI;
}
```

Solution 2: Python + C (The Python Script)

```
import ctypes, math
import numpy as np
import matplotlib.pyplot as plt

# Load the shared library
lib = ctypes.CDLL("./vector.so")

# Define C function signature
lib.angle_from_cross_c.argtypes = [ctypes.c_double, ctypes.c_double,
                                     ctypes.c_double, ctypes.c_double,
                                     ctypes.c_double]
lib.angle_from_cross_c.restype = ctypes.c_double

# Input values
a_mag, b_mag = 2.0, 7.0
cross = np.array([3.0, 2.0, 6.0])

# Call the C function
theta_deg = lib.angle_from_cross_c(a_mag, b_mag, *cross)
```

Solution 3: Pure Python

```
import math
import numpy as np
import matplotlib.pyplot as plt

# Input values
a_norm, b_norm = 2.0, 7.0
cross_ab = np.array([3.0, 2.0, 6.0])

# --- Calculation using Lagrange's Identity ---
cross_norm = np.linalg.norm(cross_ab)
dot_sq = (a_norm**2) * (b_norm**2) - cross_norm**2
dot_val = math.sqrt(dot_sq)

# --- Find angle from dot product ---
cos_theta = dot_val / (a_norm * b_norm)
theta_rad = math.acos(cos_theta)
theta_deg = math.degrees(theta_rad)
print(f"Angle is {theta_deg:.2f} degrees")
```

Plot Result

